

QUASAR: Interacting Quantum-Walk Perturbation Scanning for Blind Allosteric-Site Prediction from Protein Topology

Launch excitations at the active site of a protein's contact network, let them propagate under an interacting quantum Hamiltonian, and rank every residue by how strongly a "ligand" placed there changes what the active site feels.

Challenge: Cleveland Clinic — Unlocking undruggable targets: quantum simulation of allosteric signal propagation

Team: Entangled Minds

Paradigm: Hybrid classical-quantum; gate-based digital Hamiltonian simulation (Trotterised XXZ quantum walks) with classical coarse-graining and statistics

Platforms: AWS Braket (SV1, DM1, IonQ / IQM QPUs), Classiq; Qiskit for open reproducibility

Summary. Elastic network models (GNM/ANM) and Markov propagation models [8, 15, 16] predict allosteric coupling from contact topology alone, but they are *linear* (harmonic) or *diffusive*, and a diffusive signal forgets where it came from. We propose **QUASAR** (QUantum Allosteric Signal Analysis of Residue networks). We encode the residue contact network as an interacting hard-core-boson (XXZ) Hamiltonian, one qubit per coarse-grained node. With a single excitation and no interaction, the model reduces *exactly* to a continuous-time quantum walk on the GNM Kirchhoff matrix, the coherent counterpart of the classical diffusion models the field already uses. That gives every quantum ingredient a matched classical control. The allosteric score is a **quantum perturbation-response scan (qPRS)**: how much stiffening one residue's contacts, as a bound ligand would, changes the excitation dynamics measured at the active site. Every observable is read out in the computational basis, and excitation-number conservation provides error mitigation by post-selection at no extra circuit cost. **Preliminary evidence on the challenge targets** (Appendix A, all from apo topology): (i) on ABL1's distal myristoyl pocket, quantum-walk scores are the only topology-only scores reaching nominal significance (AUC 0.62, $p = 0.04$, uncorrected; 0.78 among residues $>12 \text{ \AA}$ from the ATP site), against 0.47 for GNM correlation and 0.54 for classical diffusion; (ii) on KRAS, *no* topology score, quantum or classical, beats simple distance to the active site (AUC 0.59); (iii) in a noisy 12-qubit emulation of our circuits (≤ 696 CNOT-equivalents), post-selection keeps the ranking at Spearman $\rho \geq 0.90$ at 1% two-qubit error, versus 0.79–0.85 without it. The linear, single-excitation walk is therefore not enough, and it is classically simulable. Phase 2 tests the core hypothesis that **interacting, multi-excitation dynamics adds predictive signal over the best matched classical control**, with decision gates, a sealed held-out set and pre-registered targets: the correct pocket in the top-5 hit list for ≥ 2 of 3 validation targets, $\Delta\text{AUC} \geq +0.05$ over the best classical control, and $\rho \geq 0.80$ between QPU and noiseless rankings at ≥ 16 qubits.

1. Problem Framing

Our reading of the problem. Given only an unbound structure, produce an $N \times N$ residue connectivity matrix and a ranked top-5 list of candidate allosteric sites, without molecular dynamics, and score it against the pocket seen in a drug-bound structure. Two things make this hard. First, the target pockets are frequently *cryptic*: the KRAS Switch-II pocket that sotorasib occupies is not formed in GDP-bound KRAS [18, 19], so static pocket detection on the apo structure [14] has nothing to find. Second, the pocket must be singled out from "random background residues and non-functional surface pockets," so a score that merely favours residues near the active site, or buried residues, is not enough. Under the challenge's elastic-network hypothesis the information must come from the dynamics that the contact topology implies. The question is which *kind* of dynamics carries the allosteric signal.

Why quantum dynamics is a credible, testable proposal. We make three claims, and each has a matched classical control (§5.3):

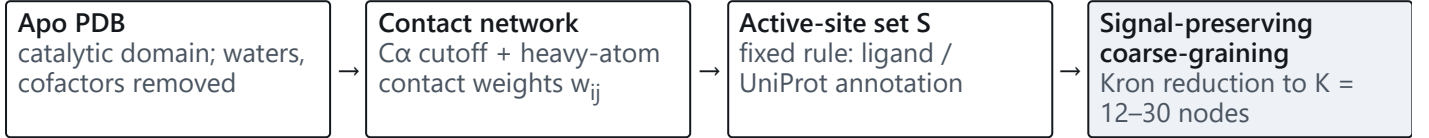
- **Coherent transport keeps memory of source and path.** Classical diffusion $e^{-\Gamma t}$ on the Kirchhoff matrix Γ relaxes to the uniform distribution, so at long range every residue looks alike. A quantum walk $e^{-i\Gamma t}$ on the *same* graph never relaxes. Its long-time transfer probability $C_{ij}^\infty = \sum_\lambda |\sum_{k \in \lambda} v_{ik} v_{jk}|^2$ is source-specific and weights residues by the vibrational modes they share with the active site, not by proximity [26, 35]. This parameter-free quantity is what ranked the distal ABL1 pocket above chance (Appendix A).
- **Interactions supply the non-linearity that classical approximations lack** [7–9]. An $\Delta \cdot n_i n_j$ term couples two excitations whenever contacting residues are displaced together, which is a minimal model of cooperative, non-harmonic contact coupling. With $M \geq 2$ excitations the dynamics is genuinely many-body: the state space grows as $C(K, M)$, and entanglement growth on non-planar contact graphs defeats tensor-network simulation at scale. This is where a quantum processor is the natural simulator [36].
- **The measurements fit the hardware.** Populations $\langle n_i \rangle$ and all two-point correlators $\langle n_i n_j \rangle$ come from a single computational-basis measurement setting, so the full connectivity matrix needs no extra bases. Because the Hamiltonian conserves excitation number, any shot with the wrong Hamming weight is a detected error and is discarded [32].

What we do not claim. We claim no demonstrated quantum advantage. At Phase 2 scales ($K \leq 30$ qubits, $M \leq 4$) every circuit is still exactly simulable classically, and we use that to verify hardware output. Our preliminary data already show that the single-excitation (linear) quantum walk fails on KRAS. The claim we will defend or refute is narrower: *interacting quantum dynamics on the contact graph ranks experimentally validated allosteric residues better than the best linear, diffusive, mean-field or geometric control, on proteins it has not seen.*

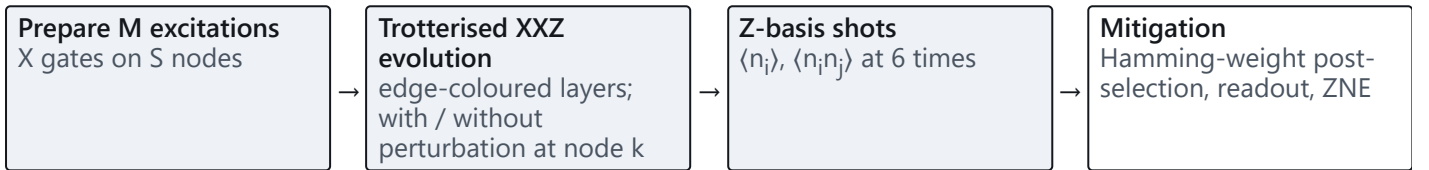
Benchmark issues we found and will raise in Week 1. From the PDB files: **6C1H** (validation, cardiac myosin) is a cryo-EM actin–myosin–ADP complex containing no mavacamten. **5TBY** (input) is a homology model of the interacting-heads motif with no nucleotide. **1OPL** (ABL1 input) already contains myristate in the target pocket, so its apo backbone is pre-organised. We will ask Cleveland Clinic to confirm the intended mavacamten-bound structure, and meanwhile we report myosin as pending (Appendix C1).

2. Technical Approach

CLASSICAL PRE-PROCESSING (CPU, seconds–minutes)



QUANTUM (AWS Braket / Classiq; classical exact simulation for verification)



CLASSICAL POST-PROCESSING & DELIVERY

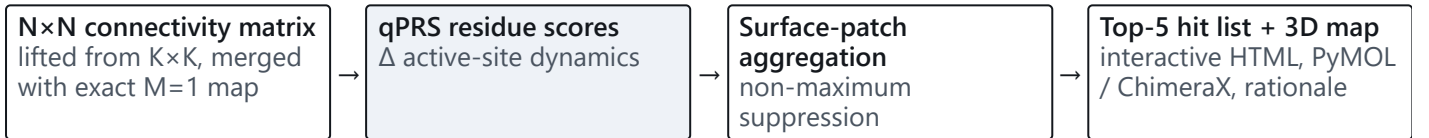


Figure 1. QUASAR pipeline. Shaded blocks are quantum or quantum-derived. The holo (drug-bound) structure is never used for scoring, only for evaluation.

2.1 From contact network to qubit Hamiltonian

Nodes are residues (or coarse-grained nodes); edges join residues whose Cα atoms lie within r_c (7.3 Å GNM default [16]), weighted by w_{ij} , the normalised heavy-atom contact count. With degrees $d_i = \sum_j w_{ij}$ we define

$$H = -\sum_{i<j} (w_{ij}/2)(X_i X_j + Y_i Y_j) + \Delta \sum_{i<j} w_{ij} n_i n_j + \sum_i d_i n_i, \quad n_i = (I - Z_i)/2.$$

In the single-excitation sector ($M = 1$), H equals the Kirchhoff matrix Γ exactly (Appendix B1), so the known physics of elastic networks is the starting point rather than an assumption. Δ and M are the only “beyond-classical” knobs. Both are tuned on a development set and then frozen.

2.2 Quantum metrics

Connectivity matrix. $C_{ij} = (1/T) \sum_t [\langle n_i n_j \rangle_t - \langle n_i \rangle_t \langle n_j \rangle_t]$ after excitations are launched from S ; for $M = 1$ this becomes the time-averaged transfer probability. **qPRS allosteric score.** A putative ligand at residue k stiffens its contacts, $w_{kj} \rightarrow (1+\epsilon)w_{kj}$, giving H_k . The score is $A_k = (1/T) \sum_t |R_k(t) - R(t)|$, where $R(t) = \sum_{s \in S} \langle n_s \rangle_t$ is the excitation retained at the active site. This is the quantum analogue of classical perturbation-response scanning [29] and applies the operational definition of allostery directly: *binding here changes function there*. **Hit list.** Residue scores are averaged over surface patches (8 Å radius around each solvent-exposed residue of the apo structure). After non-maximum suppression, the five best patches form the top-5 list.

2.3 Coarse-graining that provably keeps the topological signal

Proteins of 170–800 residues exceed near-term devices when mapped one residue per qubit. In our preliminary test, naive spectral clustering to $K = 12$ kept only $\rho = 0.43$ (KRAS) and 0.57 (ABL1) of the full-resolution ranking, which is not good enough. We will instead use **Kron reduction** [28]: keep the active-site nodes plus candidate patch centroids, and eliminate interior residues by the Schur complement $\Gamma_{\text{red}} = \Gamma_{RR} - \Gamma_{RI} \Gamma_{II}^{-1} \Gamma_{IR}$. This preserves every effective resistance (static coupling) between retained nodes *exactly*. We then prune the weakest couplings to meet a two-qubit-gate budget. A reduction is accepted only if its ranking matches the full-resolution classical simulation at $\rho \geq 0.80$, for both $M = 1$ and $M = 2$. That check is exact and affordable: the KRAS two-excitation sector has $C(169, 2) = 14,196$ states.

2.4 Circuits, depth and noise resilience

We use first-order Trotterisation with edge-coloured layers. Each coupling is one $XX+YY$ rotation (2 CNOT-equivalents), or an XXZ block (≤ 3) when $\Delta \neq 0$; on-site terms are virtual Z rotations. At $K = 12$ the preliminary KRAS/ABL1 graphs have 26/29 couplings in 9 layers. The deepest circuit (12 steps) uses 624/696 CNOT-equivalents at two-qubit depth 108, and Trotter error leaves the ranking at $\rho \geq 0.97$. Trapped-ion all-to-all connectivity (IonQ on Braket) needs no SWAPs for non-planar contact graphs. For the superconducting devices (IQM) we route and depth-optimize with Classiq synthesis. **Mitigation stack:** Hamming-weight post-selection, readout-confusion inversion, zero-noise extrapolation by gate folding [33] and Pauli twirling. **Shallow check circuit:** for $M = 1$ the evolution can also be compiled exactly into $K(K-1)/2$ Givens rotations at depth $\approx 2K$ [34], an independent low-depth test of the hardware readout.

2.5 Scaling path

Near term (Phase 2): $K \leq 24-36$ all-to-all qubits, $M \leq 4$, verified against exact simulation. **Mid term:** $K \approx 100$ residue-level nodes with $M \approx 10-25$ ($C(100,25) \approx 2 \times 10^{23}$ states), about 10^4 two-qubit gates per time point. This is beyond exact classical simulation and needs early fault-tolerant devices. **Fault tolerant:** qubitisation-based simulation, with cost growing linearly in the evolution time, over whole catalytic domains. The Phase 2 decision gates establish whether the signal justifies that path before it is pursued.

3. Feasibility and Resource Requirements

Resource	Requirement (planning estimate)	Basis
Quantum simulators	Braket SV1 / DM1 ($\leq 34 / \leq 17$ qubits); local statevector for ≤ 20 qubits	Verification and noise emulation; provided by the challenge
Quantum hardware	IonQ (all-to-all) primary, IQM secondary; $\approx 5 \times 10^5$ shots in total	$K = 16$: 6 times $\times$ 16 circuits $\times$ 1,000 shots $\approx 10^5$ per protein-configuration; ABL1 and KRAS, $M = 1$ and 2, plus calibration
Classical compute	Laptop for $M = 1$; one 64-core / 256 GB cloud node, $\approx 2,000$ core-hours, for exact $M = 2-3$ sectors	Sparse Krylov propagation, dimension $\leq 10^6$ (ABL1 $M = 2$: 101,475 states)
Software (open)	NumPy/SciPy, Braket SDK, Classiq, Qiskit, ProDy [30] (GNM/ANM/PRS), fpocket [31], FreeSASA, py3Dmol	Challenge recommends Qiskit [17]; all baselines reproducible
Data	7 challenge PDB entries (already downloaded); ≈ 25 apo/olo pairs from ASD [25]	Public; no Cleveland Clinic or proprietary data needed

Assumptions and constraints. (i) The elastic-network hypothesis holds: topology drives propagation, so there are no force fields and no MD. (ii) The active site is defined by a fixed rule decided in advance: residues within 4.5 Å of an orthosteric ligand in the input structure, otherwise UniProt catalytic annotation. (iii) Holo structures are used only for scoring. (iv) QPU access and shot allocation follow the challenge infrastructure; if allocation is tighter, hardware runs drop to $K = 12$, $M = 1$ ($\approx 1.5 \times 10^5$ shots) and the rest runs on DM1 with device-calibrated noise.

Risk	Likelihood	Mitigation / fallback
Interactions add no signal over linear or classical controls	Medium-high	Gate G2 (week 7) on the development set. If it fails, we report the negative result and deliver the best hybrid pipeline, with the proteins where coherence helps (e.g. ABL1) identified
Coarse-graining destroys the signal	Medium	Kron reduction with an exact acceptance test; larger K on 36-qubit all-to-all hardware; hardware used only for the correlator sub-blocks that need it
Over-fitting to 3 targets	High if unmanaged	Sealed ASD held-out set; KRAS and ABL1 declared “seen” because of Phase 1 exploration; frozen code hash before evaluation
Hardware noise or queue delays	Medium	Post-selection (preliminary $\rho \geq 0.90$ at 1% error); DM1 emulation calibrated to device data; $K = 12$ fallback
Benchmark ambiguity (myosin)	Known	Clarify in week 1; otherwise use a published mavacamten-bound structure and report it transparently

4. Expected Impact

What a successful PoC demonstrates. It would be the first matched-control evidence that a quantum-simulated observable, rather than a quantum label on a classical score, predicts where drugs bind allosterically, measured blind from apo topology on proteins the model has not seen. Operationally, QUASAR is the “allosteric scanner” the challenge describes. A computational-biology unit gives it a PDB file of a deprioritised target and gets back a residue probability map, a top-5 site list and an interactive 3D view in under an hour, excluding QPU queue time. Medicinal chemists can

then focus library design and screening on a handful of surface regions instead of the whole protein. Because each hit comes with its perturbation response and dominant propagation path, a chemist can see *why* a site is predicted without quantum training. **Value under a negative result:** a public benchmark that separates coherence, interaction and coarse-graining effects on allosteric prediction, a curated apo/holo evaluation set with spatially matched null models, and a reusable harness in which future quantum metrics face the same controls.

Target (pre-registered)	Success threshold	Measured on / against
Site-level hit list	Validated pocket in top 5 for ≥ 2 of 3 validation targets; $\geq 40\%$ of held-out targets (stretch 60%)	Site-hit definition in Appendix C2
Residue-level accuracy	Mean ROC-AUC ≥ 0.70 on held-out targets; $p < 0.05$ vs spatially matched null after BH correction	≥ 10 held-out ASD pairs + myosin
Quantum attribution	$\Delta\text{AUC} \geq +0.05$ over best classical control, 95% CI excluding 0	Ablation row 8 vs best of rows 1–7
Coarse-graining retention	$\rho \geq 0.80$ vs full resolution at $K \leq 24$ (preliminary naive: 0.43–0.57)	Exact classical $M = 1, 2$ simulation
Noise resilience	$\rho \geq 0.80$ mitigated QPU vs noiseless at $K \geq 16$; top-3 nodes preserved	IonQ / IQM via Braket
Circuit budget	$\leq 1,500$ CNOT-equivalents for the deepest $K = 24$ circuit	Classiq / Qiskit transpilation report
Usability	One command: PDB $\rightarrow$ $N \times N$ matrix, top-5 list, 3D HTML report	Clean-VM replication by a non-author

5. Validation Plan

5.1 Data splits and pre-registration

- **Development set** (≈ 15 ASD apo/holo pairs [25], spanning kinases, GTPases and metabolic enzymes). This is the only data used to choose r_c , T , ϵ , Δ , M , K . **Held-out set** (≥ 10 further ASD pairs + cardiac myosin): identifiers sealed in week 3 and evaluated once, after the code hash is frozen. KRAS and ABL1 were explored in Phase 1, so they are reported separately as “seen” targets. **c-Myc (1NKP)**: DNA-contacting residues serve as S ; predictions are submitted for consensus and checked for docking viability (fpocket + AutoDock Vina).
- **Statistics.** Residue-level ROC-AUC and PR-AUC; one-sided Mann–Whitney tests against a *spatially matched null* (random surface patches matched to the pocket in size and burial), because pocket residues are spatially clustered and naive residue p -values are anti-conservative. Bootstrap 95% CIs are taken over targets, with Benjamini–Hochberg correction across every metric reported. All tested metric variants are disclosed, as in Appendix A.

5.2 Timeline and decision gates (16-week sprint)

Weeks	Work	Exit criterion
1–3	Benchmark clarification; development / held-out curation (sealed); classical ladder (ProDy, fpocket, CTRW) on development set; $M = 1$ pipeline hardened	G1: preliminary results reproduced; split frozen
4–7	Interacting model ($M = 2–3$, Δ sweep), exact classical simulation; Kron coarse-graining with acceptance test	G2: row 8 beats best of rows 1–7 by $\Delta\text{AUC} \geq 0.05$ on development set, and $\rho \geq 0.80$ at $K \leq 24$. Otherwise pivot and report
8–11	Classiq synthesis; DM1 device-calibrated emulation; QPU runs at $K = 12 \rightarrow 16 \rightarrow 24$ on ABL1 and KRAS	G3: $\rho \geq 0.80$ mitigated QPU vs ideal at $K \geq 16$
12–14	Freeze; single blind held-out evaluation; c-Myc predictions and docking check; 3D visualisation report	All targets in §4 met or characterised
15–16	Clean-environment replication by a non-author; open-source release; report with resource declaration	One-command reproduction

5.3 Ablation ladder (what each row controls for)

#	Arm	Controls for
1	Distance to active site; contact degree	Trivial geometry and burial
2	fpocket static pockets on apo structure [31]	Static pocket detection (the approach the challenge aims to go beyond)
3–4	GNM cross-correlation, GNM-PRS; ANM perturbation-response scanning [29, 30]	Linear equilibrium elasticity
5	Markov propagation / commute time [8]; continuous-time random walk (same graph and times)	Classical diffusion
6	Quantum walk, $M = 1$, $\Delta = 0$ (classically exact)	Coherence and interference alone
7	Mean-field (product-state) XXZ dynamics	Non-linearity without entanglement
8	QUASAR, $M \geq 2$, $\Delta \neq 0$, full resolution (exact)	Treatment
9 / 10	QUASAR coarse-grained $K \leq 24$ (ideal) / on QPU (mitigated)	Coarse-graining loss / hardware loss

Only the gap between row 8 and the best of rows 1–7 is credited to the quantum model. The gaps 8→9 and 9→10 separate out what coarse-graining and hardware cost.

6. Hybrid / Cross-Domain Integration

Each component does what it is good at. Classical computing handles structure parsing, graph construction, coarse-graining with exact guarantees, statistics, pocket patching and visualisation. The quantum processor is used only for the interacting many-body dynamics, which is the one step whose classical cost grows combinatorially. Classical exact simulation of the same circuits at small scale is built in as a verifier, not left as an afterthought. **Integration with drug-discovery workflows:** outputs are standard artefacts: B-factor-annotated PDB files, PyMOL/ChimeraX sessions, a JSON hit list with residue indices ready for docking (e.g. AutoDock Vina box definitions), and a self-contained HTML 3D report. **AI assistance, outside the scoring path:** an optional LLM-based explainer turns each hit's propagation path into a plain-language rationale for medicinal chemists, lowering the entry barrier as the challenge requests. It never changes scores.

7. Team Capability

Member	Role on this project	Relevant expertise
Kyoung-Tae (Elijah) Kim MSc Quantum Computing, UPM (Spain)	Quantum Project Manager: sprint plan and gates G1–G3; classical ablation ladder and statistics harness; chemist-facing explainer	Project lead, voice-phishing scam detection (Microsoft); AI & ML; RAG and LangChain pipelines; agentic-AI orchestration
Naveed Qureshi MSc Quantum Computing, UPM (Spain)	Quantum Technical Lead: Hamiltonian encoding, coarse-graining under qubit budget, Trotter circuits, Classiq synthesis, Braket execution and mitigation	Quantum optimisation; satellite mission planning formulated and solved as a QUBO
Alejandro Valencia Blancas MSc Quantum Computing, UPM (Spain)	Quantum Solutions Architect: PDB/ASD data pipeline, reproducibility, 3D visualisation report, evaluation protocol and pre-registration	Software engineering and database systems; mobile communications; AI, and the ethics and epistemology of AI

Why this team. Entangled Minds formed during the MSc in Quantum Computing at Universidad Politécnica de Madrid, where we worked as a standing project team on quantum algorithms, simulation and optimisation. **Naveed** has taken a constrained operational problem and fitted it to a finite qubit budget, which is exactly the problem of coarse-graining a 450-residue protein onto 16–24 qubits without losing the signal. **Elijah** led an applied ML detection project at Microsoft, where the lesson was that a model is only as credible as its strongest baseline. He owns the classical controls, which decide whether any quantum gain is real. **Alejandro** brings the data engineering for the PDB/ASD curation and release. His training in the epistemology of AI underpins the plan's central discipline: sealed test sets, disclosed metric variants, and claims no larger than the evidence supports. The preliminary study in Appendix A is our own work. It was built for this proposal from the challenge's PDB entries, it runs end-to-end on a laptop CPU in about 20 minutes, and its results are reported whether favourable or not.

What we lack, and how we cover it. No team member is a structural biologist or medicinal chemist. We address this by relying on established, peer-reviewed tools for every biological step (ProDy, fpocket, FreeSASA, ASD annotations), by defining active sites and pocket contacts through fixed geometric rules rather than expert judgement, and by requesting a short Cleveland Clinic domain review at gates G1 and G2. **Execution readiness:** the team's AWS

account is active with Amazon Braket enabled; the PDB data and preliminary code are in hand; and all three members commit full-time (35+ hours per week) to the Phase 2 sprint.

References. [1]–[25] as numbered in the Cleveland Clinic challenge statement §7. [26] Childs, Farhi & Gutmann, Quantum Inf. Process. 1, 35 (2002) · [28] Dörfler & Bullo, IEEE Trans. Circuits Syst. I 60, 150 (2013) · [29] Atilgan & Atilgan, PLoS Comput. Biol. 5, e1000544 (2009) · [30] Bakan, Meireles & Bahar, ProDy, Bioinformatics 27, 1575 (2011) · [31] Le Guilloux, Schmidtke & Tufféry, fpocket, BMC Bioinformatics 10, 168 (2009) · [32] Bonet-Monroig et al., PRA 98, 062339 (2018) · [33] Temme, Bravyi & Gambetta, PRL 119, 180509 (2017) · [34] Kivlichan et al., PRL 120, 110501 (2018) · [35] Mülken & Blumen, Phys. Rep. 502, 37 (2011) · [36] Lloyd, Science 273, 1073 (1996). Full list in Appendix D.

Appendix — Supplementary Material

QUASAR: Interacting Quantum-Walk Perturbation Scanning for Blind Allosteric-Site Prediction from Protein Topology

Challenge: Cleveland Clinic — Quantum simulation of allosteric signal propagation **Team:** Entangled Minds
Contents: A. Preliminary results · B. Mathematical specification · C. Validation specifications · D. References

Supplements the 6-page concept proposal; § refers to proposal sections. All data are public RCSB PDB entries; all code is the team’s own (NumPy/SciPy/scikit-learn). No confidential or proprietary information is included.

A. Preliminary Results on the Challenge Targets

Setup (all rules fixed before scoring). *Input:* chain A of the apo structure, one node per residue C α , unweighted contacts within $r_c = 7.3 \text{ \AA}$. *Active site S:* residues with any heavy atom within $4.5 \text{ \AA}$ of the orthosteric ligand in the *input* file: GDP+Mg²⁺ in 4OBE (KRAS, N = 169, |S| = 23) and the ATP-site inhibitor P16 in 1OPL (ABL1, N = 451, |S| = 21). *Ground truth:* residues within $4.5 \text{ \AA}$ of the allosteric drug in the holo structure, mapped by sequence alignment and excluding S. That is sotorasib (MOV) in 6OIM, giving 16 residues at a mean C α distance of $10.5 \text{ \AA}$ from S, and asciminib (AY7) in 5MO4, giving 20 residues at $19.4 \text{ \AA}$. *Evaluation:* ROC-AUC of pocket residues against all other non-site residues, one-sided Mann–Whitney p (uncorrected), and the number of pocket residues among the top-5 ranked residues. Quantum scores use the exact single-excitation dynamics ($M = 1, \Delta = 0$); the interacting model is Phase 2 work. qPRS/cPRS use $\varepsilon = 0.5$; τ is time in units of $1/\text{mean degree}$.

Score	Type	KRAS AUC (p)	KRAS top-5	ABL1 AUC (p)	ABL1 distal-only AUC†	ABL1 AUC, $r_c=8 \text{ \AA}$
Distance to active site	geometry	0.589 (0.12)	1	0.516 (0.40)	0.694	0.516
Contact degree	geometry	0.461 (0.70)	0	0.530 (0.33)	0.545	0.510
GNM cross-correlation with S	linear elastic	0.365 (0.96)	0	0.474 (0.65)	—	0.460
GNM perturbation response (MSF at S)	linear elastic	0.537 (0.32)	2	0.423 (0.88)	0.551	0.447
Commute time to S	diffusive	0.388 (0.93)	1	0.600 (0.065)	—	—
Random walk transfer, $\tau = 10$	diffusive	0.455 (0.72)	0	0.539 (0.28)	—	0.545
Random walk perturbation (cPRS), $\tau = 10$	diffusive	0.501 (0.49)	2	0.538 (0.28)	0.723	0.544
Quantum walk transfer, $\tau = 10$	quantum	0.385 (0.93)	0	0.555 (0.20)	—	0.579
Quantum walk transfer, $\tau \rightarrow \infty$ (C $^\infty$)	quantum	0.314 (0.99)	0	0.619 (0.036)	—	0.601
qPRS, $\tau = 10$	quantum	0.465 (0.68)	0	0.555 (0.20)	0.744	0.575
qPRS, $\tau \rightarrow \infty$	quantum	0.356 (0.97)	1	0.618 (0.037)	0.775	0.592

† Pocket vs other residues $>12 \text{ \AA}$ from S; all 20 ABL1 pocket residues are distal. No score placed an ABL1 pocket residue in its top 5. Random scoring gives AUC 0.50. **Disclosure:** across both scripts we evaluated 32 score/parameter variants at up to 4 cut-offs (7.0/7.3/8.0/10 $\text{\AA}$), all in `prelim_results.json` and `qprs_results.json`. No p-value here survives multiple-comparison correction, so we treat these results as hypothesis-generating only.

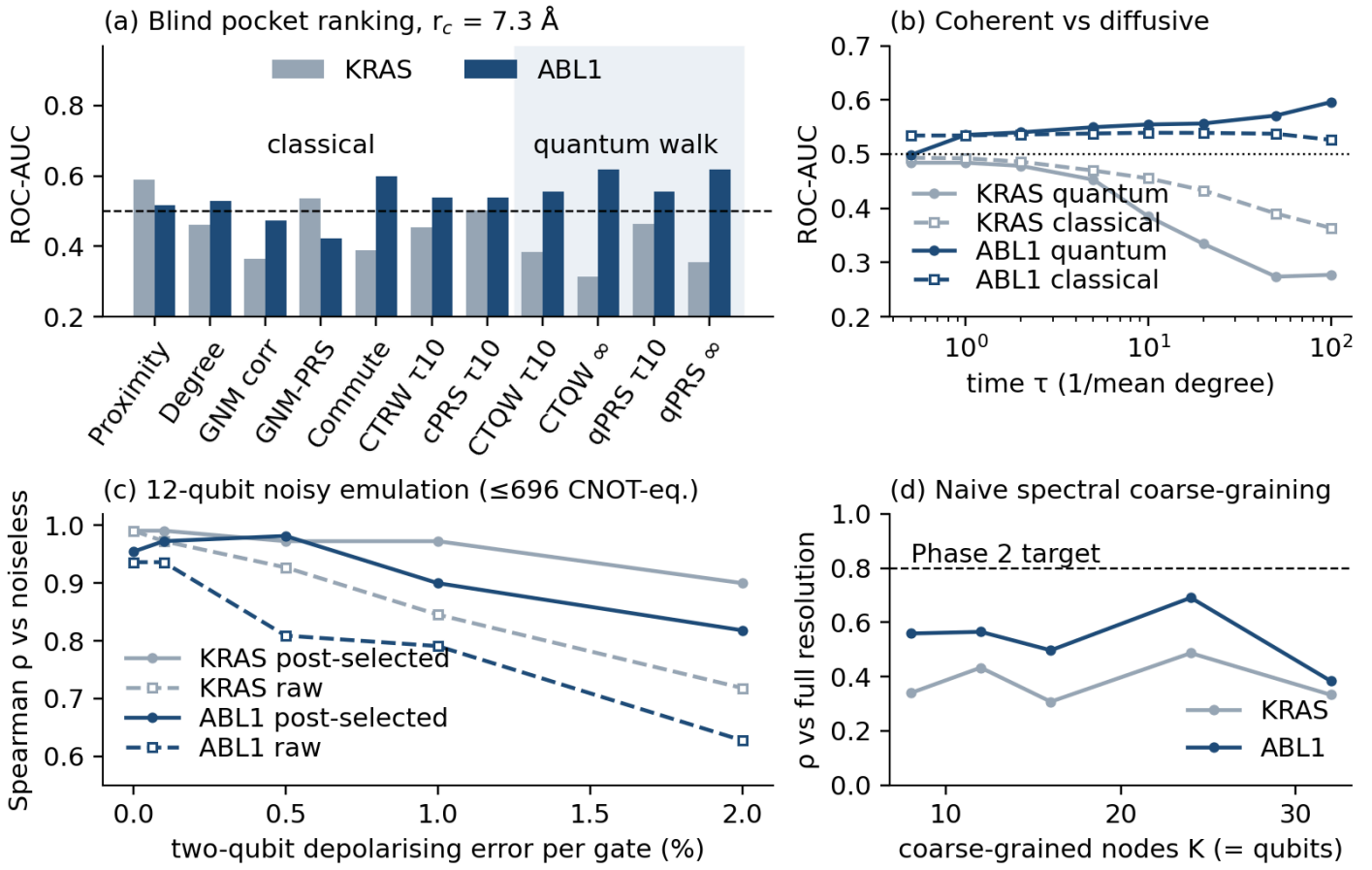


Figure A1. (a) Blind pocket-ranking AUC by score. (b) The quantum walk gains with evolution time on ABL1's distal pocket but loses on KRAS's near-site pocket; classical diffusion on the same graph never exceeds AUC 0.54. (c) Noisy emulation of the $K = 12$ circuits: rank agreement with the noiseless connectivity row, raw vs Hamming-weight post-selected. (d) Naive spectral coarse-graining keeps too little of the full-resolution ranking, which motivates Kron reduction (§2.3).

Reading the results. (1) Coherence helps most for the *distal* pocket: on ABL1 the quantum scores rise with evolution time, while diffusion on the same graph stays near chance. (2) On KRAS the Switch-II pocket lies beside the nucleotide site, so distance to the active site wins and single-excitation coherence hurts. Coherent transport alone is therefore not enough, which motivates interactions ($M \geq 2$, $\Delta \neq 0$) and patch-level scoring. (3) At $M = 1$ qPRS closely tracks the transfer score; we expect them to diverge once interactions are added.

$K = 12$ emulation; two-qubit error $p_{2q} \rightarrow$	0	0.1%	0.5%	1%	2%
KRAS	0.99 / 0.99 / 89%	0.97 / 0.99 / 78%	0.93 / 0.97 / 49%	0.85 / 0.97 / 32%	0.72 / 0.90 / 16%
ABL1	0.94 / 0.95 / 89%	0.94 / 0.97 / 77%	0.81 / 0.98 / 45%	0.79 / 0.90 / 29%	0.63 / 0.82 / 14%

Each cell: p raw / p post-selected / acceptance, with 1% readout error, 4,000 shots and 300 trajectories per circuit. Circuits: KRAS/ABL1 have 26/29 couplings in 9 colour layers, giving 52/58 CNOT-equivalents per Trotter step, with 6 time points of 2–12 steps ($\Delta t = 0.25$). The deepest circuit is 624/696 CNOT-equivalents at two-qubit depth 108. Trotterised vs exact ranking: $p = 0.98/0.97$. With post-selection the top-3 nodes are preserved at every noise level; without it, 2 of 3 are preserved at $\geq 1\%$. At $K = 16/24/32$ the couplings number 46/61/81 (KRAS) and 38/67/84 (ABL1). The noise model includes only depolarising and readout errors; coherent errors, crosstalk and T_1 decay will be characterised on the devices in Phase 2.

B. Mathematical Specification

B1. Single-excitation sector equals the Kirchhoff matrix. Let $|a\rangle = X_a|0\dots 0\rangle$. Since $(X_a X_b + Y_a Y_b)/2 = \sigma_a^+ \sigma_b^- + \sigma_a^- \sigma_b^+$, we have $\langle a|H|b\rangle = -w_{ab}$ for $a \neq b$ and $\langle a|H|a\rangle = d_a$, while the $n_i n_j$ term vanishes. Hence $H|_{M=1} = D - W = \Gamma$. Every term commutes with $\hat{N} = \sum n_i$, so M is conserved. A readout bitstring with Hamming weight $\neq M$ therefore certifies an error, and discarding it is exact symmetry verification [32].

B2. Long-time limits. With $\Gamma = \sum_k \lambda_k v_k v_k^T$, the quantum walk gives $(1/T) \int_0^T |\langle j|e^{-i\Gamma t}|i\rangle|^2 dt = \sum_{k,1} v_{ik} v_{jk} v_{i1} v_{j1} \text{sinc}((\lambda_k - \lambda_1)T)$, which tends to $\sum_{\lambda} |\sum_{k \in \lambda} v_{ik} v_{jk}|^2$, a source-dependent limit. The classical walk gives $(1/T) \int_0^T e^{-\Gamma t} dt = \sum_k v_k v_k^T (1 - e^{-\lambda_k T})/(\lambda_k T)$, which tends to $11^T/N$ for a connected graph and is therefore uninformative. We used these closed forms for the exact scores in Appendix A.

B3. Trotter step and gate cost. $e^{-iH\Delta t} \approx \prod_{\text{layers}} \prod_{(a,b)} \exp[i\theta_{ab}(X_a X_b + Y_a Y_b)/2 - i\phi_{ab} Z_a Z_b] \cdot \prod_a R_Z$, where $\theta_{ab} = w_{ab}\Delta t$ and $\phi_{ab} = \Delta w_{ab}\Delta t/4$. Couplings sharing no qubit commute and form one layer, so a greedy edge colouring uses $\leq d_{\max}+1$ layers. With $\phi = 0$ the XY block is locally equivalent to iSWAP⁰ (2 CNOTs); the general XXZ block needs ≤ 3 CNOTs.

B4. Kron reduction. Partition the nodes into retained R and interior I. Then $\Gamma_{\text{red}} = \Gamma_{RR} - \Gamma_{RI}\Gamma_{II}^{-1}\Gamma_{IR}$ is again a Laplacian with non-negative couplings, and the effective resistance $(e_a - e_b)^T \Gamma^+ (e_a - e_b)$ is identical in the full and reduced graphs for every $a, b \in R$ [28]. Retained set: all S-cluster representatives plus up to $K - |S_K|$ surface-patch centroids chosen by farthest-point sampling; residues inherit the score of their nearest retained patch.

C. Validation Specifications

C1. Benchmark inspection (from the PDB files)

Target	Input (apo)	Validation (holo)	What we found	Action
KRAS G12C	4OBE: GDP-bound KRAS, 2 chains	6OIM: sotorasib (MOV) + GDP	Switch-II pocket absent in input, as expected for a cryptic site; 16 pocket residues	Chain A; used in Appendix A
BCR-ABL1	1OPL: SH3-SH2-kinase, P16 + myristate	5MO4: asciminib (AY7) + nilotinib	Myristate already occupies the target pocket in the input	Topology only (HETATM ignored); caveat reported
Cardiac myosin	5TBY: homology model, interacting-heads motif	6C1H: actin– myosin–ADP cryo-EM	No mavacamten in 6C1H; no nucleotide in 5TBY	Confirm intended holo with Cleveland Clinic in week 1
c-Myc (open)	1NKP: Myc/Max bHLHZip on DNA	None (consensus + docking)	Short, largely disordered chains; DNA defines the functional site	S = DNA-contacting residues; docking check

C2. Metric definitions (fixed before Phase 2 results)

- **Pocket residues:** any heavy atom within 4.5 Å of the allosteric ligand in holo, mapped to input numbering by sequence alignment. **Site hit:** a predicted patch counts as a hit if its centroid lies within 8 Å of the ligand centroid *and* it shares ≥ 3 residues with the pocket. **Top-5 hit rate:** fraction of targets with ≥ 1 hit among the five predicted sites.
- **Spatially matched null:** 1,000 random surface patches with the same residue count and mean relative SASA ($\pm 10\%$) as the true pocket; the empirical p-value is the fraction of null patches scoring $\geq$ the true pocket. **Retention:** Spearman ρ between full-resolution and coarse-grained (or noisy) residue rankings over non-site residues.
- **Statistics:** bootstrap over targets (10,000 resamples) for 95% CIs on mean AUC and ΔAUC ; Benjamini–Hochberg at $q = 0.05$ across all reported metrics.

D. References

[1]–[25] As numbered in the Cleveland Clinic challenge statement, §7 (incl. [7] Stock & Hamm 2018; [8] Chennubhotla & Bahar 2007; [15] Das et al. 2014; [16] Erman 2006; [17] Javadi-Abhari et al. 2024; [18] Ostrem et al. 2013; [19] Canon et al. 2019; [20] Wylie et al. 2017; [25] Shen et al. 2016).

[26] A. M. Childs, E. Farhi, S. Gutmann. An example of the difference between quantum and classical random walks. *Quantum Inf. Process.* 1, 35–43 (2002).

[28] F. Dörfler, F. Bullo. Kron reduction of graphs with applications to electrical networks. *IEEE Trans. Circuits Syst. I* 60(1), 150–163 (2013).

[29] C. Atilgan, A. R. Atilgan. Perturbation-response scanning reveals ligand entry-exit mechanisms of ferric binding protein. *PLoS Comput. Biol.* 5, e1000544 (2009).

[30] A. Bakan, L. M. Meireles, I. Bahar. ProDy: protein dynamics inferred from theory and experiments. *Bioinformatics* 27, 1575–1577 (2011).

[31] V. Le Guilloux, P. Schmidtke, P. Tufféry. Fpocket: an open source platform for ligand pocket detection. *BMC Bioinformatics* 10, 168 (2009).

[32] X. Bonet-Monroig, R. Sagastizabal, M. Singh, T. E. O’Brien. Low-cost error mitigation by symmetry verification. *Phys. Rev. A* 98, 062339 (2018).

[33] K. Temme, S. Bravyi, J. M. Gambetta. Error mitigation for short-depth quantum circuits. *Phys. Rev. Lett.* 119, 180509 (2017).

[34] I. D. Kivlichan et al. Quantum simulation of electronic structure with linear depth and connectivity. *Phys. Rev. Lett.* 120, 110501 (2018).

[35] O. Mülken, A. Blumen. Continuous-time quantum walks: models for coherent transport on complex networks. *Phys. Rep.* 502, 37–87 (2011).

[36] S. Lloyd. Universal quantum simulators. *Science* 273, 1073–1078 (1996).

Software and data: RCSB PDB entries 4OBE, 6OIM, 1OPL, 5MO4, 5TBY, 6C1H, 1NKP; NumPy, SciPy, scikit-learn, Matplotlib. Preliminary scripts: `allo_prelim.py`, `qprs.py`, `noise_emul.py`, released with the Phase 2 repository.